# WeiboDRP: Sentiment Analysis on Chinese Social Media with Multi-Level Comment Structures

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**Abstract.** Social media provides valuable insights into public sentiment, but analyzing sentiment accurately remains challenging due to the widespread presence of multi-level comment structures (MLCS) in social media texts. Existing research lacks dedicated datasets and systematic evaluation metrics for MLCS-based sentiment analysis tasks. This study proposes WeiboDRP, a large-scale dataset explicitly designed for Chinese sentiment analysis with MLCS annotation. Benchmark evaluations using dictionary methods, fine-tuned pre-trained models, and zero-shot large language models on this dataset reveal that DeepSeek-R1 achieves the best overall performance. Additionally, this study proposes a novel evaluation metric called the Hierarchical Consistency Score (HCS), which effectively measures models' ability to capture and utilize hierarchical contextual information in MLCS. The effectiveness of HCS is verified through ablation experiments. Its cross-domain generalizability is further demonstrated on a publicly available benchmark dataset. These findings highlight the importance of effectively understanding and leveraging the contextual information provided by MLCS to improve classification performance in social media-based sentiment analysis tasks.

**Keywords:** Social media · Sentiment Analysis · Hierarchical Comment Structure · Argument Quality Assessment · Large Language Model

## 1 Introduction

Social media platforms have become central to public discourse, allowing individuals to express opinions and share views [1]. In China, Weibo is among the largest social media platforms, with over 588 million active users by December 2024 [22]. Sentiment analysis of Weibo posts provides valuable insights into public attitudes toward various topics. However, accurate sentiment classification on Chinese social media remains difficult. Firstly, Chinese social media texts are informal, noisy, and linguistically complex [21]. Secondly, social media comments often present multi-level comment structures (MLCS), consisting of a main post followed by multiple levels of replies. Ignoring hierarchical context can lead models to classify sentiment incorrectly [19]. For example, a positive reply to a negative main post might reinforce the negative sentiment. Without context, models might mistakenly label it as positive.

Previous studies have proposed various methods to enhance context utilization in sentiment analysis. These studies demonstrate that incorporating hierarchical structures can significantly improve sentiment classification [8, 11]. Methods include hierarchical BiLSTM with attention [8], graph-based modelling, multi-level encoding, and user history integration [11, 19]. But these methods primarily rely on artificial or dialogue-based datasets like DailyDialog and MultiWOZ, which differ significantly from real-world social media interactions [2, 17]. Meanwhile, existing studies typically use standard metrics such as accuracy and macro-F1, lacking specialized metrics for evaluating context dependency across comment levels [2, 24]. This limitation hampers the comparability and understanding of models in hierarchical sentiment analysis tasks.

To address these gaps, we introduce WeiboDRP, a large-scale Chinese dataset explicitly designed for sentiment analysis with MLCS annotations. This dataset comprises approximately 180,000 Weibo posts, including 6,882 complete MLCS groups. A carefully annotated subset, WeiboDRP-MLCS, contains 1,738 manually labeled posts validated through statistical measures. We evaluate various sentiment analysis approaches, including dictionary-based methods, fine-tuned pre-trained language models (PLMs), and large language models (LLMs). We propose a novel metric, the Hierarchical Consistency Score (HCS), to assess how effectively models utilize hierarchical sentiment contexts specifically. This metric systematically measures how well models use context information across levels in MLCS. We validated its effectiveness through ablation studies and validated the cross-domain generalizability of HCS on an open-access dataset: Pheme-Rumour-Scheme dataset. The main contributions of this paper are as follows:

- A large-scale dataset designed explicitly for sentiment analysis of MLCS in Chinese social media, including a manually annotated subset.
- A set of baseline experiments evaluating the crossing of different sentiment analysis methods on the dataset, providing a comprehensive benchmark.
- A novel evaluation metric, Hierarchical Consistency Score (HCS), was proposed to assess the utilization of hierarchical context by models.

All datasets, the annotation codebook, code, and model parameters in this study are publicly available in the Appendix A.

## 2 Dataset

In this section, we will explain the WeiboDRP dataset in detail, starting with the definition of MLCS, the construction of the dataset, the data preprocessing, and the manual annotation process.

### 2.1 Multi-Level Comment Structures

Social media posts can typically be categorized into three hierarchical levels, main post and the corresponding level 1 and level 2 comments. A main post and

**Table 1.** Example of Multi-Level Comment Structures

| | | |
|---|---|---|
| **Main Post** | (a) This policy is seriously hurting our social welfare. We're getting less pension because of it. | |
| **Level 1 Comment** | (b) Exactly! Couldn't have said it better. | (c) Hmm, I'm not so sure about that. In the end, I think they'll factor in the actual conditions. |
| **Level 2 Comment** | (d) Preach! | (e) Hope so. |

all its associated comments form one complete Multi-Level Comment Structure (MLCS) group. In MLCS sentiment analysis, models must consider parent comments to classify child comments correctly. As the example shown in Table 1, if the model analyzes comment (b) in isolation, it may be misclassified as positive due to its positive tone. However, combined with the negative sentiment context of its parent comment (a), the model can accurately identify that comment (b) expresses approval of negative emotions and should therefore be classified as negative.

Contextual information is most effective within direct parent-child comment pairs. For example, when analyzing comment (c), the model can use the main post (a) context. When analyzing the level 2 comment (e), the model can use the direct Level 1 comment (c) as context. However, combining the Main Post (a) with a Level 2 comment (d) outside this direct relationship may lead to a context mismatch, which can cause errors in sentiment classification.

The concept of an MLCS Pair (or Parent-Child Pair) is defined as formalizing this idea. An MLCS pair refers to a direct reply relationship between the two comments.

### 2.2 Dataset Construction

**Raw Data Acquisition.** This study uses 'delayed retirement' as a keyword and crawls 128 days of Weibo data from the Weibo platform from July 7, 2024 to November 12, 2024. In the initial stage, 15,111 Main Post were captured. Subsequently, using the unique identifier (mid) of the main post as the index, all the corresponding Level 1 Comment 116,551 and Level 2 Comment 64,659 were further captured. Finally, the raw dataset containing 196,221 original text data was formed.

**Data Preprocessing.** A text preprocessing pipeline was designed for Chinese social media data to ensure high-quality input for the sentiment analysis model. Noise specific to Weibo was removed. This included forwarding links (//@), topic tags (#...#), user mentions (@username), and hyperlinks (http(s)://). Zero-width characters and non-Chinese and non-English characters were removed. Only Chinese, English, numbers, and basic punctuation marks were retained. Continuous blank spaces were standardized to maintain a consistent text format. Baidu's open-source LAC tool performed word segmentation and part-of-speech tagging [9].

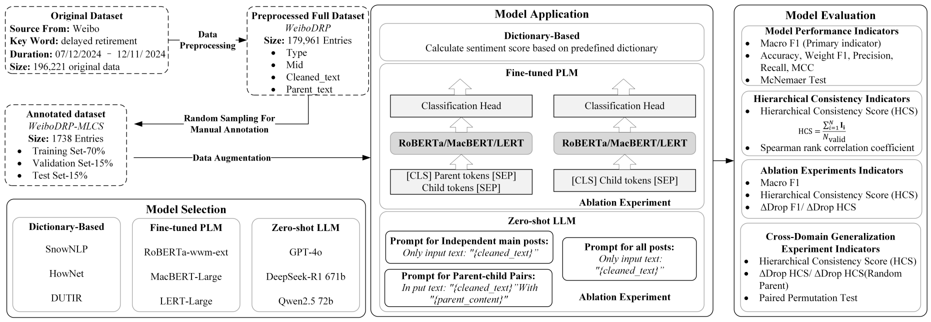


**Fig. 1.** System Workflow

Only four semantically rich parts of speech were retained: nouns, verbs, adjectives, and idioms. Stop word filtering was applied using a custom stop word list. This list combined function words and symbols from the HowNet lexicon with other common Chinese function words and punctuation marks [13].

After preprocessing, the WeiboDRP dataset was constructed. The 'Type' field indicates the hierarchical structure, while the 'cleaned text' and 'parent content' fields represent the direct parent-child relationships. The final dataset contains 179,961 valid text entries, including 6,626 complete MLCS Groups and 82,567 Parent-Child Pairs.

**Manual Annotation.** Due to the large volume, a subset named WeiboDRP-MLCS was randomly sampled from WeiboDRP for manual annotation. It includes 1738 entries with 576 complete MLCS groups, 1088 parent–child pairs, and 68 isolated main posts. We implemented a prudent pipeline, detailed steps and the codebook shown in Appendix A. Three initial annotators were selected from native Chinese speakers with diverse ages, academic backgrounds, and jobs. The adjudicator holds a postdoctoral degree in Chinese linguistics to balance subjective bias. All participants receive codebook training. We start formal labeling only after a 100-sample calibration achieves Fleiss' Kappa $> 0.70$. The three annotators independently label two categories, sentiment polarity (Positive, Neutral, Negative) and attitude tendency (Agree, Neutral, Disagree). Then, the ground truth is set by majority vote, and the adjudicator resolves ties. Fleiss' Kappa was calculated to evaluate annotation reliability, 0.705 for sentiment and 0.704 for attitude, with bootstrap confidence intervals and permutation tests showing $p < 0.001$.

The final label distribution was as follows: sentiment polarity (Negative 880, Neutral 689, Positive 169) and attitude tendency (Agree 388, Disagree 220, Neutral 1130). The annotated is divided into training (70%), validation (15%), and test (15%) sets, and we keep each MLCS group within a single split to preserve hierarchy.

## 3 Methodology

Fig. 1 shows the methodological framework in detail, covering model selection, application procedures, and evaluation strategies. The framework also incorporates ablation and cross-domain experiments aimed at evaluating the effectiveness and generalizability of Hierarchical Consistency Score (HCS).

### 3.1 Model Selection

Based on the background, this study evaluate three sentiment method types: dictionary-based baselines, fine-tuned pre-trained language models (PLMs), and zero-shot large language models (LLMs). For the baselines, we use three Chinese sentiment lexicons, including SnowNLP [24], HowNet [20], and DUTIR [3]. They are simple and highly interpretable, and help clarify the performance of traditional approaches in sentiment analysis of social media content [21]. For PLMs, we fine-tune RoBERTa-wwm-ext, MacBERT, and LERT [5, 21]. We choose them for their stable and widely verified performance on Chinese text. For LLMs, we test GPT-4o, DeepSeek-R1, and Qwen2.5, which show strong zero-shot sentiment ability and solid Chinese understanding [21]. This selection enables a clear comparison between classic lexicons, supervised PLMs, and zero-shot LLMs within a unified evaluation framework.

### 3.2 Model Application

For lexicon baselines, SnowNLP computes a sentiment score, HowNet counts positive and negative words, DUTIR outputs seven categories, and we map them to positive, neutral, and negative. For PLMs, in order to address class imbalance in the training set, a widely used data augmentation pipeline has been used, including character insertion, character deletion, and BERT-based synonym replacement [18]. Then, we apply the same pipeline to fine-tune three PLMs with a context-aware design. The input embedding layer jointly encoded parent and child comment pairs to capture hierarchical context. See Appendix A for details of the training strategy. For LLMs, we run zero-shot LLMs with prompt-based inference that highlights MLCS context, we design different templates for each text type, as shown in Fig. 2.

### 3.3 Model Evaluation

This study evaluates and compares the performance of various sentiment analysis methods using conventional metrics and the proposed HCS. In addition, ablation experiments were carried out to validate the effectiveness of HCS, and a cross-domain evaluation on a publicly accessible benchmark dataset was conducted to assess the generalizability of the MLCS framework and HCS.

**Performance Indicator Selection.** This study uses Macro F1 as the primary evaluation metric. It treats all classes equally, balances precision and recall,

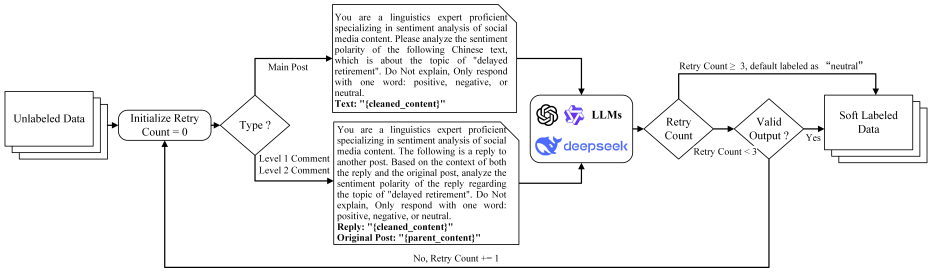


**Fig. 2.** Zero-shot LLMs Data Labeling Workflow

and supports fair comparisons across different model architectures. This also ensures that minority classes are not overlooked [10]. The following auxiliary metrics are reported for reference and statistical tests: accuracy, weighted F1, Matthew's correlation coefficient (MCC), precision, recall, McNemar test, and Spearman correlation [4, 7, 14]. By combining these indicators, the evaluation framework measures overall classification performance and quantifies structural consistency in multi-level social media interactions.

**Hierarchical Consistency Score.** To evaluate a model's ability to capture parent-child context in MLCS, this study introduces the Hierarchical Consistency Score (HCS). This metric quantifies whether sentiment predictions align with the annotated attitude between Parent-Child Pairs.

Let the set of all child comments be $\{C_1, C_2, \ldots, C_N\}$, each annotated with an attitude label $attitude_i \in \{\text{Agree}, \text{Disagree}, \text{Neutral}\}$. Let $\hat{y}_{parent}$ and $\hat{y}_{child}$ denote the sentiment polarity for the parent and child posts. The expected consistency rule is defined as:

- If $\hat{y}_{parent} = \hat{y}_{child}$, then $attitude_i = \text{Agree}$.
- If $\hat{y}_{parent} \neq \hat{y}_{child}$, then $attitude_i = \text{Disagree}$.
- If a child post does not rely on its parent for contextual interpretation (e.g., topic drift), or the post is a Main Post with no parent, then $attitude_i = \text{Neutral}$.

An indicator function $\mathbf{I}(\cdot)$ represents whether the expectation is satisfied:

$$\mathbf{I}_i = \begin{cases} 1, & \text{if the expected parent–child relationship satisfied} \\ 0, & \text{Otherwise} \end{cases} \tag{1}$$

The HCS is calculated as:

$$\text{HCS} = \frac{\sum_{i=1}^{N} \mathbf{I}_i}{N_{valid}} \tag{2}$$

Where $N_{valid}$ is the number of child comments labeled as 'Agree' or 'Disagree'.

HCS offers four advantages. First, it measures context sensitivity by testing parent–child alignment. Higher HCS means better use of context. Second, it supports error diagnosis beyond accuracy, since a model may label each post

**Table 2.** Model Performance Comparison

| Method | Accuracy | Macro F1 | Weighted F1 | MCC | Precision | Recall | HCS |
|---|---|---|---|---|---|---|---|
| SnowNLP | 0.3511 | 0.3192 | 0.3903 | 0.1349 | 0.4195 | 0.4587 | 0.5185 |
| HowNet | 0.3588 | 0.3455 | 0.4143 | 0.1475 | 0.4414 | 0.4327 | 0.4537 |
| DUTIR | 0.4046 | 0.3744 | 0.4078 | 0.1791 | 0.4615 | 0.4931 | 0.4259 |
| RoBERTa-wwm-ext | 0.6641 | 0.5990 | 0.6863 | 0.4579 | 0.5966 | 0.6748 | 0.6574 |
| MacBERT-Large | 0.7176 | 0.6533 | 0.5183 | 0.5183 | 0.6356 | 0.7103 | 0.6944 |
| LERT-Large | 0.7634 | 0.6988 | 0.5886 | 0.5886 | 0.6891 | **0.7165** | 0.6667 |
| GPT-4o | 0.7824 | 0.7089 | 0.7803 | 0.5987 | 0.7388 | 0.6898 | 0.6944 |
| DeepSeek-R1 671b | **0.8053** | **0.7184** | **0.7968** | **0.6443** | **0.7563** | 0.7023 | **0.7685** |
| Qwen2.5 72b | 0.7366 | 0.6698 | 0.7447 | 0.5559 | 0.6612 | 0.6967 | 0.6204 |
| Ground Trueth Label | – | – | – | – | – | – | 0.9815 |

correctly but still miss their relationship. Third, it enables fair comparison with single-text baselines by quantifying the gain from hierarchical information. Finally, it has practical value: models with higher HCS better trace sentiment flow and discourse alignment across threads.

**Ablation Experiments.** We run an ablation to test the effect of MLCS context and to validate HCS. The first goal is to compare model performance with and without the MLCS context. The second goal is to test whether HCS measures a model's use of that context. We evaluate two model types, fine-tuned PLMs and zero-shot LLMs. We exclude dictionary methods because they inherently cannot use hierarchical context. We compare two inputs, a context-aware setting that encodes parent–child pairs, and a child only variant that encodes only the child comment.

**Cross-Domain Generalization Experiment.** We test cross-lingual, cross-platform, and cross-topic generalization of MLCS and HCS on the Pheme-Rumour-Scheme dataset [12]. This publicly available dataset contains 4842 Twitter posts in English and German across nine topics. Each post has a hierarchy label and a human attitude label[3]. We evaluate three inputs: Parent+Child, Child Only, and Random Parent+Child. The third pair each child with a random parent from the same topic to break the actual context and test whether HCS gains come from meaningful context or just extra text. We use a zero-shot setting, keep the same prompts, and compute HCS for the three LLMs. We include only zero-shot LLMs here, since the three PLMs were pre-trained on Chinese corpora and may not handle English or German well.

## 4 Result and Analysis

This section reports all models' performance on WeiboDRP-MLCS and tests statistical significance to verify robustness.

[3] Pheme-Rumour-Scheme dataset: https://www.kaggle.com/datasets/usharengaraju/pheme-dataset.

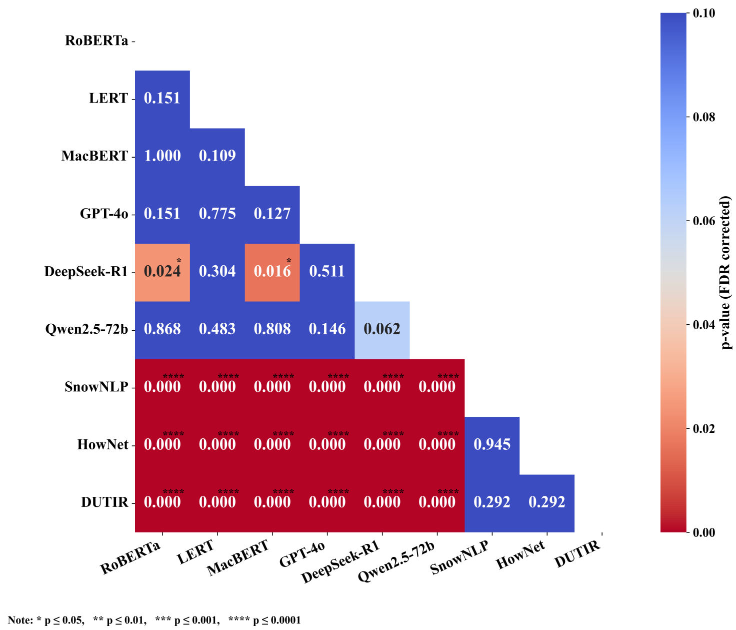


**Fig. 3.** McNemar Test p-values (FDR)

### 4.1 Model Performance

Table 2 reports the comparisons of all models. Dictionary-based methods perform close to the random baseline(0.33) and the lowest HCS, which aligns with the theoretical expectation of their mechanisms. PLMs achieve higher performance, indicating better use of hierarchical structure. Among LLMs, DeepSeek-R1 achieves the best results on almost all metrics, showing that the inclusion of hierarchical context yields the most significant performance gains.

The theoretical upper bound for HCS is 1.0. However, the observed HCS for the ground-truth labels is 0.9815. This slight gap may arise from the inherent ambiguity of sentiments near class boundaries under a coarse-grained scheme. As the sample size grows and the sentiment categories become more fine-grained, the human-annotated HCS is expected to approach its theoretical maximum asymptotically.

Figure 3 presents the pairwise McNemar test results for all models, with p-values adjusted by the Benjamini–Hochberg procedure to control the false discovery rate (FDR) [15]. These results provide statistical evidence of performance differences across models. All dictionary-based methods show statistically significant gaps when compared with others. No significant differences are observed among PLMs, indicating comparable performance despite architectural differences. Within LLMs, Qwen2.5 (72B) shows a borderline significant difference versus DeepSeek-R1 (671B), suggesting that, under the same prompt design,

**Table 3.** Ablation Experiment Results

| Model | Input | Macro F1 | HCS |
|---|---|---|---|
| **RoBERTa-wwm-ext** | Parent+Child | 0.5990 | 0.6574 |
| | Child Only | 0.5866 | 0.6019 |
| | $\Delta$ (Drop) | **-0.0124** | **-0.0555** |
| **MacBERT-Large** | Parent+Child | 0.6533 | 0.6944 |
| | Child Only | 0.6036 | 0.6759 |
| | $\Delta$ (Drop) | **-0.0497** | **-0.0185** |
| **LERT-Large** | Parent+Child | 0.6988 | 0.6667 |
| | Child Only | 0.6250 | 0.5648 |
| | $\Delta$ (Drop) | **-0.0738** | **-0.1019** |
| **GPT-4o** | Parent+Child | 0.7089 | 0.6944 |
| | Child Only | 0.6836 | 0.6389 |
| | $\Delta$ (Drop) | **-0.0253** | **-0.0555** |
| **DeepSeek-R1 671b** | Parent+Child | 0.7184 | 0.7685 |
| | Child Only | 0.6348 | 0.7130 |
| | $\Delta$ (Drop) | **-0.0836** | **-0.0555** |
| **Qwen2.5 72b** | Parent+Child | 0.6698 | 0.6204 |
| | Child Only | 0.6736 | 0.6296 |
| | $\Delta$ (Drop) | 0.0038 | 0.0092 |

LLMs with different parameter scales may process hierarchical information differently.

### 4.2 Ablation Experiment Result

Table 3 reports an ablation that removes parent comments from six context-aware models. Macro F1 drops for all models except Qwen2.5-72B. The most significant drop is on DeepSeek-R1, followed by LERT, which shows the value of hierarchical context. HCS shows similar patterns. DeepSeek-R1 (671B) and GPT-4o (∼200B) drop equally, suggesting large-scale LLMs rely on hierarchy to a similar degree. For Qwen2.5-72B, the positive $\Delta$ suggests that appending parent text does not yield a net benefit. Recent studies show that LLMs can be distracted by appended context, even when semantically related, through an over-reasoning mechanism that diverts the model from the core target [16, 23]. Among PLMs, LERT-Large falls the most, possibly due to its semantically oriented pretraining, which makes it more sensitive to disruptions in contextual alignment, leading it to be more reliant on MLCS information. Overall, explicit hierarchy improves performance, and HCS captures this dependence. Spearman's $\rho = 0.828$ ($p < 0.01$) between Macro F1 and HCS confirms the construct validity of HCS [6].

### 4.3 Cross-Damian Generalization Experiment Result

Table 4 reports cross-domain results for three LLMs under three inputs on the publicly available Pheme-Rumour-Scheme dataset [12]. We build parent–child pairs from the topic and hierarchy tags and map each pair to the attitude label. With the same prompts as in Fig. 2, and add a role setting 'You are a linguistics expert proficient in English and German' to improve cross-lingual reliability and reduce bias. We compute HCS and its Drop, run 2000 paired permutation

**Table 4.** Cross-Domain Generalization Experiment Result

| Model | Fleiss' K (Sentiment) | Input | HCS | 95% CI | p-value |
|---|---|---|---|---|---|
| **GPT-4o** | 0.6383 | Parent + Child | 0.5622 | – | – |
| | | Child Only | 0.5112 | – | – |
| | | $\Delta$ Drop | **-0.0510** | [0.0245, 0.0795] | **0.0003** |
| | | Random Parent + Child | 0.3956 | – | – |
| | | $\Delta$ Drop (Random Parent) | **-0.1156** | [-0.1462, -0.0850] | **0.0000** |
| **DeepSeek-R1** | 0.6853 | Parent + Child | 0.5629 | – | – |
| | | Child Only | 0.5105 | – | – |
| | | $\Delta$ Drop | **-0.0524** | [0.0286, 0.0755] | **0.0000** |
| | | Random Parent + Child | 0.3855 | – | – |
| | | $\Delta$ Drop (Random Parent) | **-0.1250** | [-0.1543, -0.0979] | **0.0000** |
| **Qwen2.5 72b** | 0.6003 | Parent + Child | 0.5370 | – | – |
| | | Child Only | 0.5370 | – | – |
| | | $\Delta$ Drop | 0.0000 | [-0.0231, 0.0218] | 1.0000 |
| | | Random Parent + Child | 0.3916 | – | – |
| | | $\Delta$ Drop (Random Parent) | **-0.1454** | [-0.1720, -0.1203] | **0.0000** |

tests, and report 95% bootstrap confidence intervals. Fleiss' Kappa shows that all three LLMs demonstrated high predictive agreement in sentiment outputs across different input configurations. In the true parent–child setting, GPT-4o and DeepSeek-R1 show significant HCS drops, while Qwen2.5-72B shows no change, which aligns with the WeiboDRP ablation results. In the random parent setting, HCS drops increase significantly (127%–158%) and are statistically significant. Qwen2.5 also drops, suggesting that Qwen2.5 is more vulnerable to semantic noise introduced by incorrect hierarchy inputs. This aligns with the ablation finding in Table 3, where Qwen2.5-72B shows no benefit from true hierarchical context, further supporting the contextual distraction interpretation. These results align with the main findings: HCS reacts to meaningful hierarchical context rather than input length or model bias. It further demonstrates the utility of MLCS modeling and the robustness and generalizability of the proposed HCS metric.

## 5 Conclusion and Limitations

This study introduces WeiboDRP, a large-scale Chinese social media dataset with Multi-level Comment Structures (MLCS). Using this dataset, we compare different sentiment analysis methods. The results show that large language models outperform fine-tuned and dictionary-based models. DeepSeek-R1 achieves the best overall performance on both macro-averaged and HCS metrics.

The experiments confirm that hierarchical context is important for understanding online discussions. Models that include parent–child context show higher accuracy and consistency. We propose the Hierarchical Consistency Score (HCS) to measure how well models capture this context. The ablation tests support its effectiveness and show that models rely on context in different ways.

However, this study has some limitations. First, it focuses on a single policy topic, which may introduce semantic bias; future work should incorporate diverse topics to improve generalization. Second, it covers only short comment threads, while real discussions can be deeper and more complex. Third, it uses a three-class sentiment scheme that cannot capture subtle emotions. In particular, sarcasm in MLCS often inverts the surface polarity of a child comment; under the current scheme, HCS treats such cases as sentiment disagreement, which is logically correct given the annotation but does not distinguish genuine disagreement from sarcastic agreement. Incorporating sarcasm-aware labels or a finer-grained attitude taxonomy could enable HCS to differentiate these cases. Future work will address deeper structures, broader topics, and richer emotion modelling.

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**Disclosure of Interests.** The authors declare that they have no competing interests.

## A Code and Data Availability

All datasets, annotation codebook, model parameters and computational resources usage are available at https://github.com/WillisDin/WeiboDRP.

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